DATA ANALYSIS PROJECT REPORT

Project Title	Hotel Booking Cancellation Analysis
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1. Executive Summary

This report analyzes hotel booking cancellation patterns using a Kaggle dataset of 119,390 hotel reservations across City Hotels and Resort Hotels from 2015 to 2017. The primary goal is to identify the key variables that drive cancellation behavior and to propose data-driven strategies to reduce cancellation rates. After cleaning and preprocessing the data, several critical findings emerged: the overall cancellation rate was 27%, with City Hotels experiencing a significantly higher rate (30%) compared to Resort Hotels (23.7%). Most bookings are made in advance, with Long-Term and Mid-Term bookings dominating (~69%). Last-minute bookings are rare, suggesting customers prefer early reservation planning. Cancellation behavior is heavily influenced by deposit policy. Non-refundable bookings show extremely high cancellation rates (~94.7%), indicating either high booking instability or policy-related cancellations. In contrast, refundable and no-deposit bookings maintain significantly lower and more stable cancellation rates (~24–27%).

2. Introduction and Project Objectives

Hotel reservation cancellations represent a major source of revenue uncertainty in the hospitality industry. When guests cancel bookings – especially at short notice – hotels face unsold inventory that cannot be reallocated. Understanding which guest segments, booking channels, and timing patterns are most prone to cancellation enables hotels to design smarter pricing, deposit, and communication policies.

Item	Details
Dataset name	Hotel Booking Demand (Kaggle)
Dataset source	https://www.kaggle.com/datasets/ahmedsafwatgb20/hotel-bookingscsv?resource=download
Main project goal	Identify cancellation drivers and propose reduction strategies
Research Question 1	What are the key variables influencing hotel reservation cancellations?
Research Question 2	What actionable strategies can be proposed to reduce cancellation rates?

3. Dataset Description

The dataset contains one row per hotel booking and covers both City Hotel and Resort Hotel reservations. Each booking records the guest demographics, booking channel, stay characteristics, financial metrics (Average Daily Rate), and cancellation outcome. The dataset spans three years (2015–2017) and provides a rich set of 32 features enabling multivariate analysis.

3.1 Basic Dataset Information

Property	Value
Number of rows	119,390
Number of columns	32
Data types present	Numerical, Categorical, Date-time
Target or key variable(s)	is_canceled (binary: yes / no)
Potential issues observed at first glance	agent, company columns had extensive missingness

3.2 Initial Observations

Upon initial inspection, the dataset contained several data quality issues. The agent and company columns were predominantly missing, indicating sparse tracking of booking intermediaries. The Average Daily Rate (ADR) contained extreme outliers, with one record exceeding €5,000 per night.

4. Data Cleaning and Preparation

The dataset required several cleaning steps before analysis. Extreme outliers in the ADR column were removed by applying a threshold of 5,000 to eliminate unrealistic entries. Columns with excessive missing values (agent and company) were dropped since they could not be reliably imputed and were not essential to the research. were handled by removing affected rows, and duplicate records were also eliminated to ensure accurate results.

Issue Found	Action Taken	Reason / Justification	Code Reference
ADR extreme outliers	Removed rows with adr ≥ 5,000	Single-night rates >5,000 are unrealistic; likely data entry errors	Cell 9
agent, company (missing values)	Dropped both columns entirely	High missingness rate makes imputation unreliable; not core to research questions	Cell 10
Remaining NaN values across columns	Dropped rows with any NaN	Ensures clean numeric operations	Cell 11
Duplicate booking records (n=31,984)	Removed with drop_duplicates()	Prevents double-counting in aggregations and cancellation rates	Cell 13

5. Feature Engineering

Several new features were created to improve analysis. “Season” groups months into categories to reveal tourism trends, “is_family” classifies bookings for behavioral comparison, and “lead_time_category” segments

booking time to study its effect on cancellations. These transformations simplify the data and make patterns easier to interpret.

New Feature	How It Was Created	Why It Is Useful
lead_time_category	IF x < 7 → Last_Minute ELSE IF x < 30 → Short_Term ELSE IF x < 90 → Mid_Term ELSE → Long_Term	Converts continuous lead time into interpretable risk groups for cancellation analysis
season	get_season() mapping arrival_date_month to Spring/Summer/Fall/Winter	Captures seasonal demand patterns and cancellation fluctuation across the year
is_family	IF adults > 2 → group ELSE IF adults > 2 AND (children > 0 OR babies > 0) → full_family ELSE IF adults ≤ 2 AND (children > 0 OR babies > 0) → family ELSE IF adults = 2 → double ELSE → single	Identifies family bookings which show different cancellation behavior than solo/couple travelers

6. Analysis Methodology

The analysis was conducted using a structured data science workflow to explore hotel booking behavior and identify factors influencing reservation cancellations.

pandas was used as the primary tool for data handling. It was applied to load the dataset, clean missing values, create new derived features such as season, is_family, and lead_time_category, and perform grouping and aggregation operations to summarize patterns in cancellations and reservations.

NumPy was used for numerical and statistical computations, including calculation of cancellation rates, proportions, and custom metrics. It also supported condition-based calculations for deriving additional analytical insights.

Matplotlib was used for data visualization. Various plots such as bar charts, histograms, scatter plots, and time-series graphs were created to visually interpret relationships between variables like season, lead time, pricing (ADR), and cancellation behavior.

Seaborn was used for heatmap for visualization pattern.

Overall, this combination of tools enabled efficient data preprocessing, statistical analysis, and visual interpretation to answer the research questions effectively.

7. Analysis and Visualizations

7.1 Research Question 1

What are the key variables influencing hotel reservation cancellations?

The analysis identified fourmajor categories of variables that significantly influence cancellation behavior: lead time, deposit policy, booking channel (market segment), hotel type. The table below summarizes cancellation rates across key dimensions:

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Variable	**Category**	**Cancellation Rate**	**Key Insight**
Hotel Type	City Hotel	30%	City hotels face more cancellations than resort hotels
Hotel Type	Resort Hotel	23.7%	Lower cancellation rate; guests more committed to leisure trips
Deposit Type	Non Refund	94.7%	Paradoxically near-universal cancellation despite non-refundable policy
Deposit Type	No Deposit	26.8%	Majority of bookings; moderate cancellation rate
Deposit Type	Refundable	24.3%	Majority of bookings; moderate cancellation rate
Market Segment	Groups	27%	High volume + elevated cancellation rate
Market Segment	Online TA	35%	Highest cancellation risk segment
Market Segment	Corporate	12.3%	Low cancellation; business travelers honor commitments
Lead Time	Very Long (90+ days)	36%	Plans change more often for bookings far in advance
Lead Time	Short (0-6 days)	8.16%	Lowest cancellation; imminent travel = committed guest

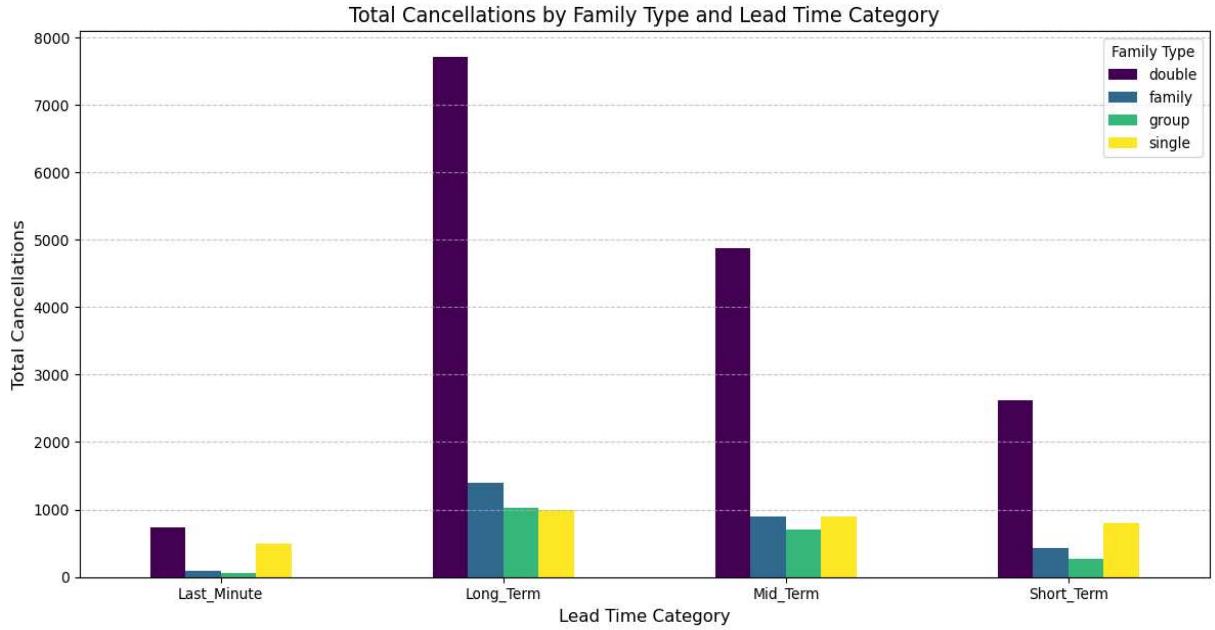

Figure 1. Total cancellations categorized by family type and lead time, illustrating a dominant trend of cancellations among "double" travelers, which scales significantly as the booking lead time increases.

Figure 1 illustrates the data reveals that "double" travelers are significantly more likely to cancel than any other group, particularly for bookings made far in advance where cancellations peak at nearly 8,000. Across all categories, there is a clear positive correlation between lead time and cancellation volume, suggesting that as the window between booking and the stay increases, the likelihood of a plan falling through rises drastically. While "single" travelers show more stability across short and mid-term lead times, the "family" and "group" types consistently maintain the lowest cancellation counts, indicating these segments may be more committed to their travel dates once booked.

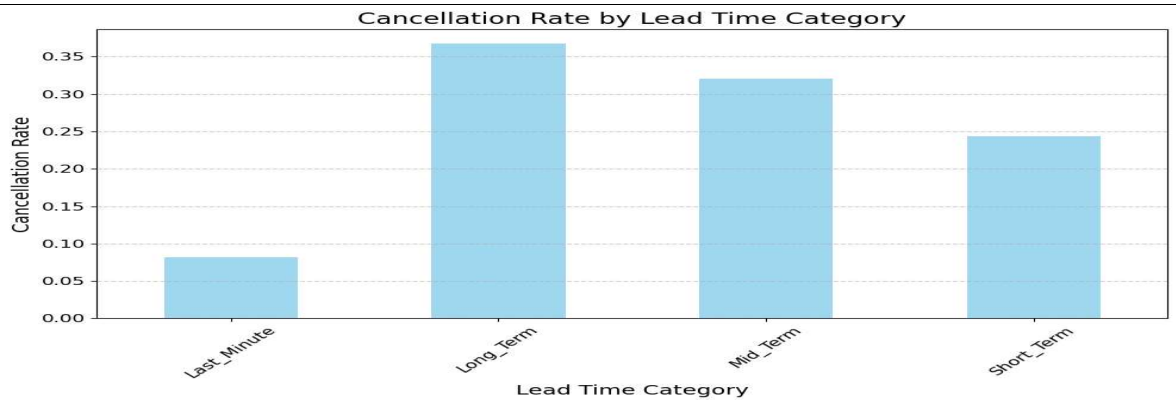

Figure 2. Cancellation rates across different lead time categories, demonstrating a direct relationship between advance booking duration and the probability of cancellation.

Figure 2 indicates a strong positive correlation between lead time and the likelihood of cancellation, with "Long_Term" bookings reaching a peak rate of approximately 37%. In contrast, "Last_Minute" bookings exhibit the highest commitment level, with a cancellation rate below 10%, which is roughly four times lower than the long-term peak. This pattern suggests that as the time horizon for a trip increases, the probability of schedule changes or external factors interfering with travel plans rises significantly.

7.2 Research Question 2

What actionable strategies can be proposed to reduce cancellation rates?

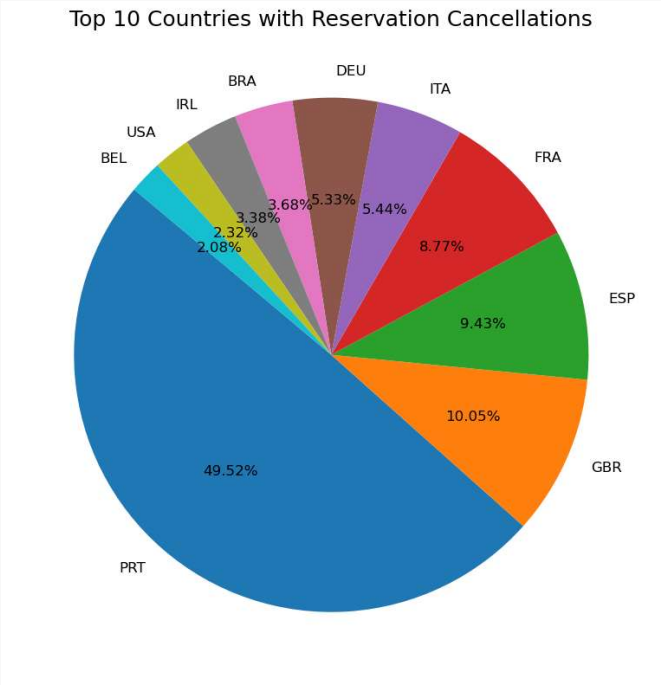

Figure 3. Distribution of reservation cancellations among the top 10 countries, highlighting a significant geographical concentration in Portugal.

Figure 3 shows a massive disparity in cancellation origins, with Portugal (PRT) accounting for nearly half (49.52%) of all cancellations among the top 10 countries. This single-country volume is nearly five times higher than that of the next highest country, Great Britain (GBR) at 10.05%The remaining eight countries, including major markets like Spain and France, contribute relatively small individual shares, suggesting that the cancellation issue is heavily localized or specific to the regional booking patterns within Portugal.

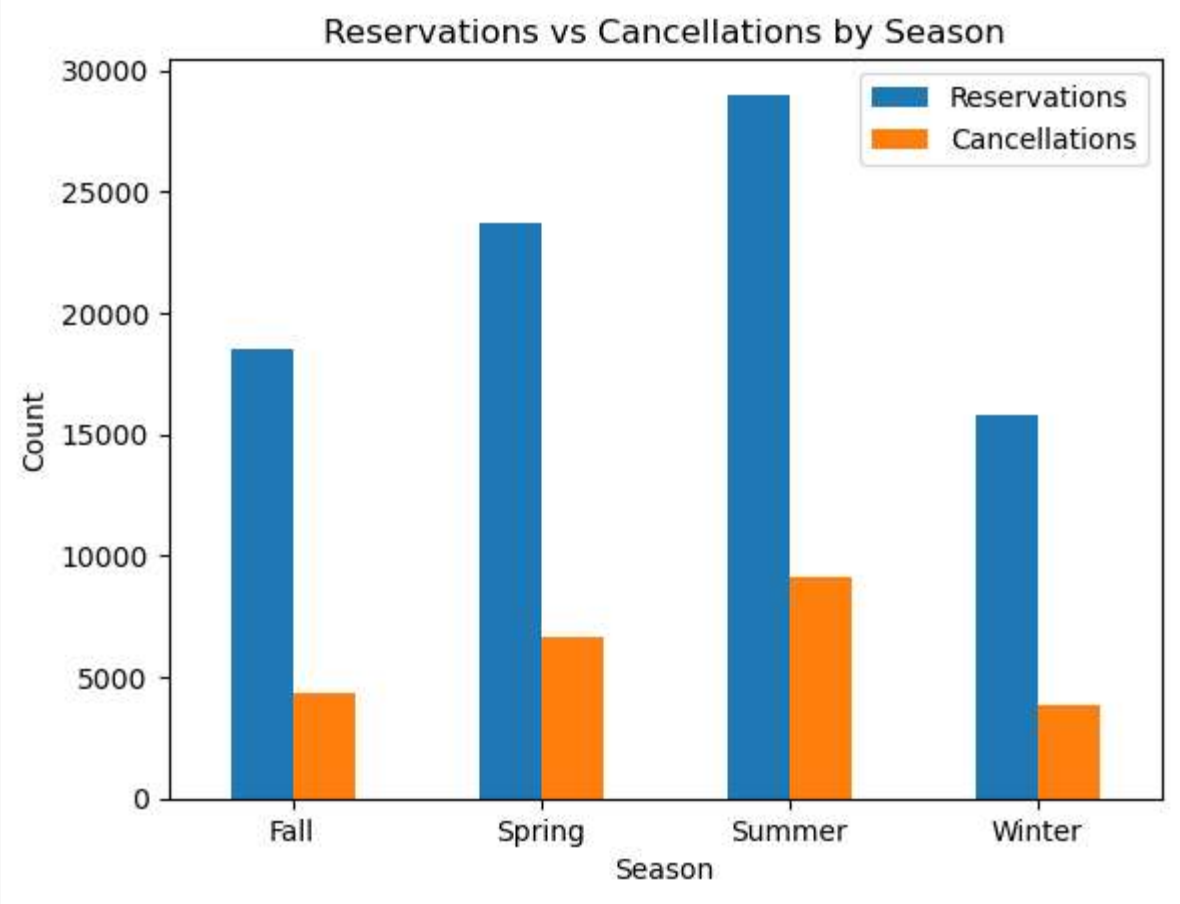

Figure 4. Comparison of total reservations and cancellations across the four seasons, highlighting the seasonal peak in activity during the summer months.

Figure indicates that summer is the peak season for travel activity, yielding nearly 30,000 reservations and a corresponding high of approximately 9,000 cancellations. While the total volume of both metrics fluctuates—dropping to their lowest levels in winter—the ratio between cancellations and reservations remains relatively consistent across all seasons. This pattern suggests that while demand is highly seasonal, the underlying factors driving cancellation behavior are stable throughout the year.

8. NumPy-Based Custom Computation

Component	Description
Computation used	Custom Cancellation Risk Score using NumPy weighted combination
Purpose	Produce a single interpretable score per booking to flag high-risk reservations before they cancel
Formula or logic	Risk = 0.5 × lead_time_norm + 0.3 × prev_cancel_flag + 0.2 × no_special_request_flag

Result / takeaway	Canceled bookings have a higher average risk score (0.207 vs 0.143).

9. Key Findings

[Write at least 5 evidence-based findings from your analysis. Avoid vague statements such as “the graph is nice” or “the values are high.”]

1. Finding 1 — Lead Time is the Strongest Predictor: Cancellation rate increases monotonically from 8.16% for bookings made 0–6 days in advance to 36% for bookings made more than 90 days in advance.
2. Finding 2 — Non-Refundable Deposits Backfire: Bookings with a Non Refund deposit policy cancel at 94.7%.
3. Finding 3 — City Hotels Face Structural Cancellation Risk: City Hotels cancel at 30% vs. 23.7%for Resort Hotels.
4. Finding 4 — Group bookings have a much higher cancellation rate (34.25%) compared to single bookings (19.64%), indicating that group reservations are significantly more likely to be canceled.
5. Finding 5 — Geographic Concentration Creates Targeted Opportunity: Portugal (PRT) alone accounts for 49.52%+ of all cancellations in the dataset.

10. Limitations

While the analysis identifies links between booking details and cancellations, it cannot prove that one actually causes the other, as hidden factors like the season might influence both. The study is also limited because significant data regarding travel agents and corporate bookings had to be removed, potentially overlooking important business patterns.

Additionally, because the data is from 2015 to 2017, it may not reflect modern habits influenced by the pandemic or newer online travel sites. The extremely high cancellation rate for "non-refundable" deposits likely points to a data error rather than actual customer behavior, so it should be viewed with caution. Finally, the results heavily represent Portugal, meaning the findings might not apply perfectly to other parts of the world.

11. Conclusion

This project analyzed 119,390 hotel booking records to determine that lead time, deposit policy, and geographic origin are the primary drivers of reservation cancellations. The analysis revealed that long-term bookings and the non-refundable deposit category have the highest propensity for cancellation, with Portugal serving as the dominant geographic source of these lost reservations. To stabilize revenue, hotels should implement automated re-confirmation prompts for long-lead bookings, redesign non-refundable policies to include partial-refund tiers, and develop localized outreach strategies for high-volume markets like Portugal, the UK, and Spain. Additionally, creating pre-stay engagement touchpoints for Online Travel Agent (OTA) and group segments could reduce risks, while future machine learning models could use these findings to predict cancellation probabilities in real-time.

12. References / Dataset Source

Mostipak, J. (2020). Hotel Booking Demand [Dataset]. Kaggle.
https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand

Original paper: Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41–49.